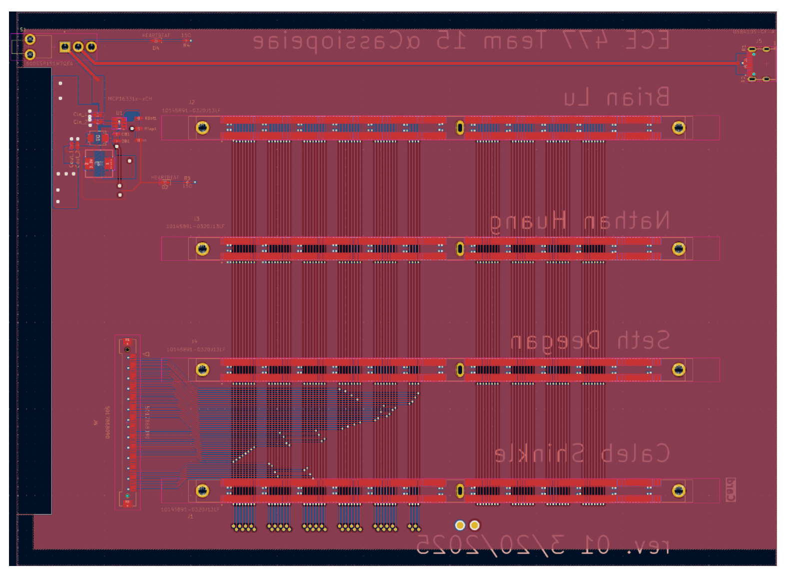
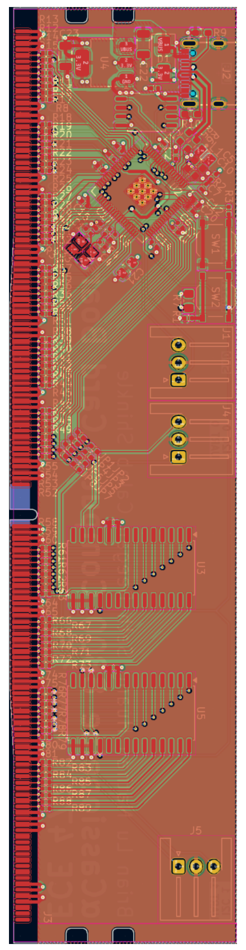
αCASSIOPEIAE 8800

ELECTRICAL

The αCassiopeiae 8800 operates on a modern, efficient power system. A standard 5V/3A USB-C power supply provides input to the Backplane PCB. Onboard, a regulated step-down buck converter efficiently transforms this input to the 3.3V required by all system components. Power is distributed from the Backplane to the Multi-function Cards (CPU, RAM, etc.) via the DDR4 DIMM slots and to the Front Panel PCB through flat flex connectors, ensuring stable voltage levels across the system.

Each core board (Front Panel, CPU, RAM, UART) utilizes a Raspberry Pi RP2350B microcontroller, supported by an external crystal oscillator and 128MB of external flash memory.

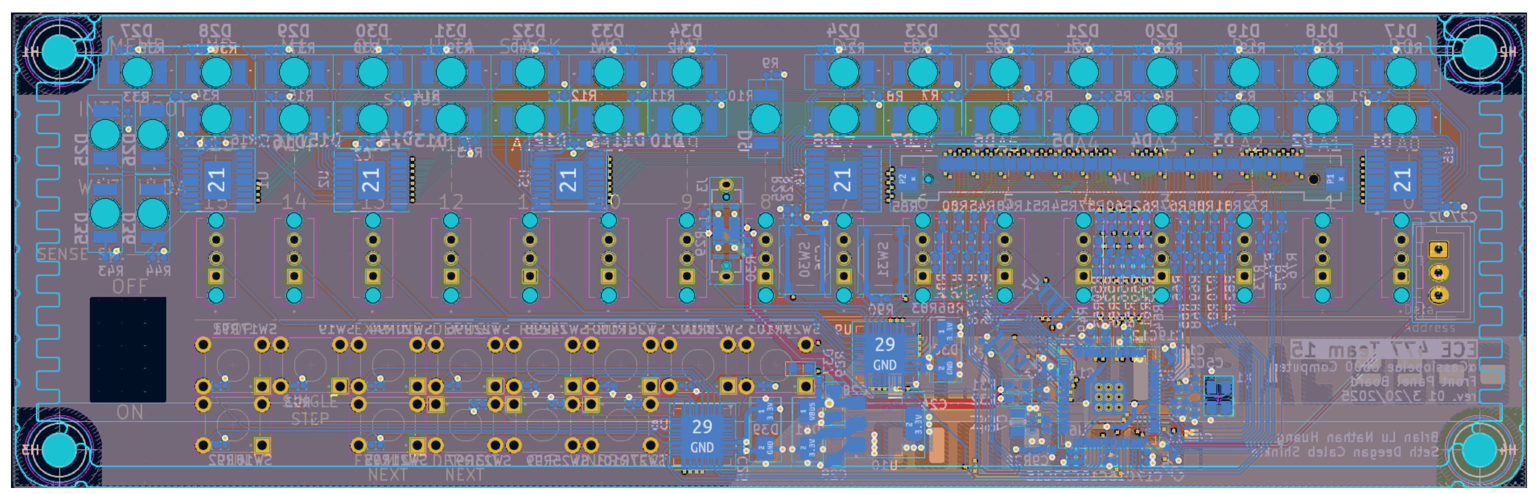




When MITS launched the Altair 8800 in 1975 it ignited the personal-computer revolution, introducing the S-100 bus, front-panel programming switches, and a price that let hobbyists own a real computer.

Our αCassiopeiae 8800 reimagines that landmark machine in a form-factor 2.8 time smaller while preserving its hands-on spirit. Implemented with multiple Raspberry Pi RP2350B microcontrollers emulates the Intel 8080 CPU, RAM, and front panel cycle-accurately, drives the classic switch-and-LED interface, and exposes an open 44-signal back-plane for add-on cards.

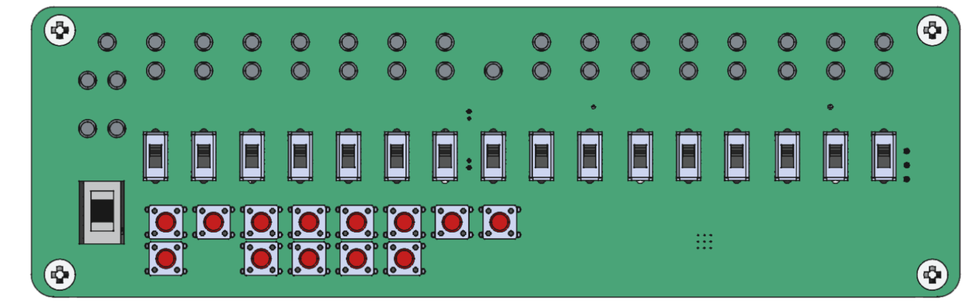
Powered from a single USB-C supply and running modern Rust firmware, αCass 8800 offers the tactile learning experience of the Altair with the convenience and hackability of 2025 electronics, an homage to the computer that started it all.



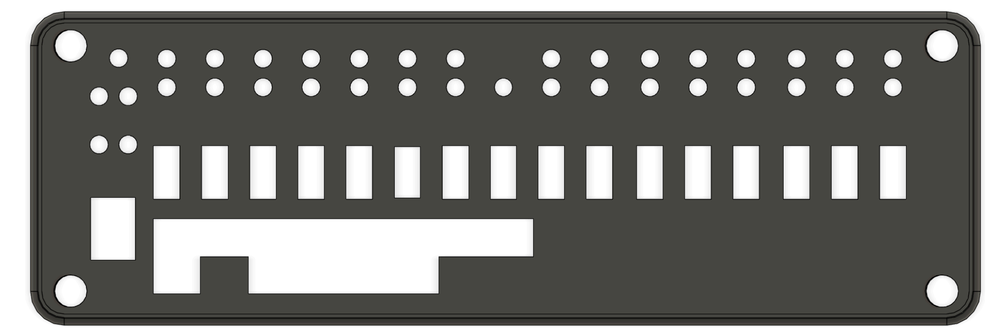
MECHANICAL

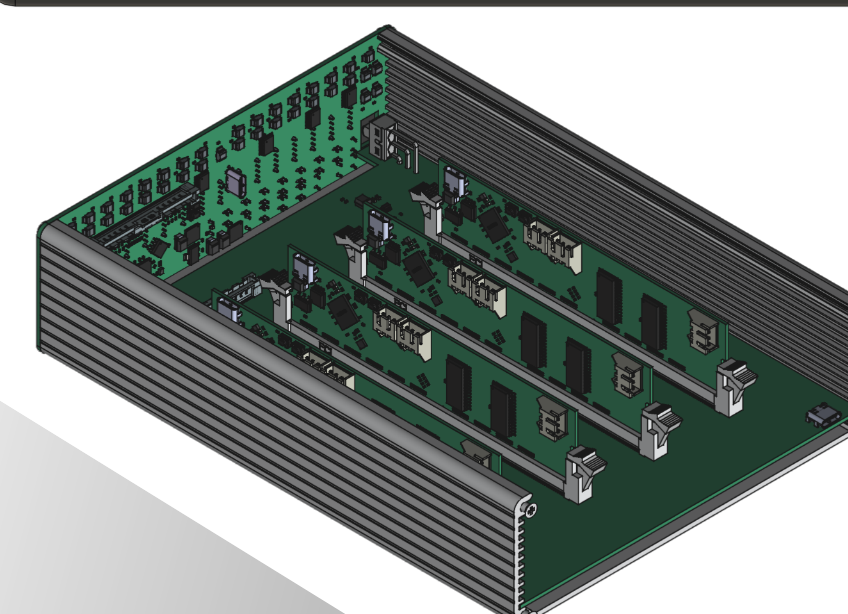
A two-piece extruded-aluminum chassis slides apart for service yet echoes the original Altair silhouette.

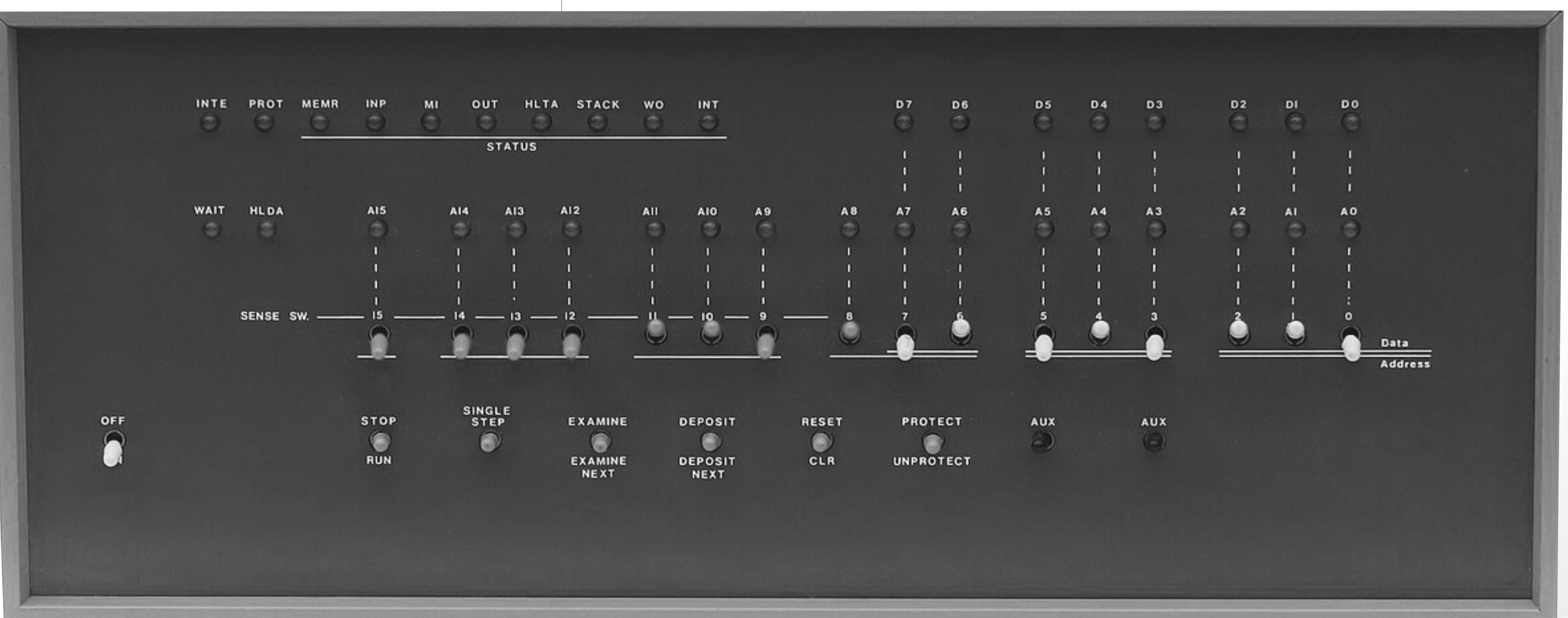
The Front-Panel PCB doubles as the face-plate, secured by four machine screws; its solder-mask and silkscreen recreate the vintage aesthetic.



Internally, the Back-plane rides on the enclosure's guide ribs, and cards plug vertically into DDR4 sockets, no wiring harnesses required. Flat-flex cables maintain a low profile between panel and back-plane







2.8x SMALLER